

System Requirements Analysis (SRA)

**Project Name: LabTrack - University Laboratory
Equipment Management System**

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1. Introduction

1.1 Purpose

This document specifies the software requirements for LabTrack, a web-based university laboratory equipment management system. It serves as a formal agreement between the developers and stakeholders on what the system shall do, ensuring all functional and non-functional requirements are testable and unambiguous.

1.2 Scope

LabTrack handles the lifecycle of laboratory equipment management. The system shall allow students to browse, reserve, collect, and return equipment. It shall enable Technical Officers to process reservations, issue equipment, and log maintenance, while allowing Administrators to manage organizational structures and personnel.

1.3 Objectives

- To digitize and streamline the equipment reservation and borrowing process.
- To maintain accurate, real-time inventory tracking for laboratory equipment.
- To automate penalty calculations for overdue equipment and process test payments.

1.4 Intended Audience

This document is intended for project evaluators, software developers, database administrators, and system testers involved in the LabTrack project.

1.5 Definitions and Abbreviations

- **JWT:** JSON Web Token, used for securely transmitting information and authentication.
- **Stripe Sandbox:** A testing environment provided by Stripe to simulate payment processing without real currency.
- **ERD:** Entity Relationship Diagram.

2. System Overview

2.1 Existing System

Universities typically rely on manual ledgers or disjointed spreadsheets to track equipment. Currently, laboratories operate entirely independently under their respective departments, siloing equipment management.

2.2 Problems with Existing System

Because laboratories operate independently, it is highly difficult and time-consuming for students from one department to request or borrow equipment from another department. Furthermore, manual tracking results in inaccurate stock levels, lost inventory, and a lack of transparency for students attempting to find available equipment for their projects.

- **Lost and Misplaced Equipment:** Lab staff cannot see who currently has specific tools, leading to misplaced items and costly losses.
- **Double-Booking & Schedule Clashes:** Manual logs allow overlapping bookings, causing student project delays during busy exam periods.
- **Neglected Maintenance:** Maintenance schedules are tracked manually or forgotten, leading to sudden equipment breakdowns during class labs.
- **Slow and Difficult Audits:** End-of-semester inventory checks require lab managers to spend days manually cross-checking paper logs against physical items.
- **Unassigned Damage & Fees:** Manual returns fail to record item conditions, making it hard to identify who damaged or returned tools late.

2.3 Proposed System

The proposed LabTrack system is a centralized web application featuring a React (Vite) frontend and a Node.js/Express.js backend. It utilizes PostgreSQL for robust data management and integrates with Stripe Sandbox for fine payments. By unifying the system across all departments and capturing detailed student records, administrators can trust the borrowing process and maintain concrete digital evidence to track inter-department and cross-departmental equipment sharing securely.

2.4 System Goals

- Ensure accurate, real-time tracking of available equipment stock across all departments.
- Collect and enforce overdue fines effectively through automated calculations and integrated payment processing.
- Provide a seamless, role-based interface for all stakeholders.

2.5 System Users / Stakeholders

- **Student:** Self-registered users who reserve equipment and pay fines.
- **Technical Officer:** Staff assigned to specific laboratories to manage stock, reservations, and maintenance.
- **Admin:** System administrators managing departments, laboratories, and staff accounts.

2.6 System Environment

The system shall operate over standard web browsers, hosted on a Node.js server environment, interfacing with a PostgreSQL relational database.

3. Functional Requirements

ID	Requirement	Priority	Actor
FR-001	The system shall allow anyone to register an account, assigning them the 'Student' role by default.	High	Student
FR-002	The system shall allow users to log into the system using valid credentials and authenticate via JWT.	High	All Users
FR-003	The system shall allow Administrators to create 'Technical Officer' accounts.	High	Administrator
FR-004	The system shall allow Administrators to assign a Technical Officer to a specific laboratory.	High	Administrator
FR-005	The system shall allow students to browse equipment, add items to a reservation cart, and select a date, start time, and end time.	High	Student
FR-006	The system shall verify available quantities during reservation and assign successfully submitted reservations a 'Pending' status.	High	System

FR-007	The system shall allow Technical Officers to update 'Pending' reservations to 'Approved' or 'Rejected' exclusively for their assigned laboratory.	High	Technical Officer
FR-008	The system shall create a borrowing record and change the reservation status to 'Completed' when an officer issues 'Approved' equipment.	High	System
FR-009	The system shall restore the available stock quantity when a student returns borrowed equipment.	High	System
FR-010	The system shall calculate and record an 'Unpaid' fine for each full 24-hour period an item is returned after its due time.	Medium	System
FR-011	The system shall allow students to process fine payments via the Stripe Sandbox test payment flow, marking the fine as 'Paid' upon success.	Medium	Student
FR-012	The system shall allow Technical Officers to create 'Open' maintenance tickets for faulty equipment units.	High	Technical Officer
FR-013	The system shall deduct available stock when a maintenance ticket is 'Open' and restore it when the ticket is marked 'Resolved'.	High	System

4. Non-Functional Requirements

4.1 Security

- **NFR-SEC-001:** The system shall encrypt all user passwords using the bcrypt hashing algorithm prior to storing them in the database.
- **NFR-SEC-002:** The system shall enforce role-based access control via JSON Web Tokens (JWT) to restrict administrative and officer functions to authorized personnel only.
- **NFR-SEC-003:** The system shall isolate student data, ensuring that students cannot view, edit, or interact with the reservation records, borrowing histories, or fine details of other students.

4.2 Performance

- **NFR-PER-001:** The system shall return equipment search results and inventory availability statuses within 2 seconds under normal operating conditions.
- **NFR-PER-002:** The system shall dynamically calculate available stock in real-time without noticeable latency when processing concurrent reservation requests.

4.3 Data Integrity

- **NFR-DI-001:** The system shall enforce logical stock limits, ensuring that an equipment item's available quantity never evaluates to a negative number.

4.4 Usability

- **NFR-USE-001:** The system shall feature a responsive user interface that adapts seamlessly to desktop, tablet, and mobile displays to ensure cross-device accessibility.
- **NFR-USE-002:** The system shall provide clear, actionable validation messages when users submit invalid inputs, such as attempting to reserve equipment that exceeds available stock.

4.5 Scalability

- **NFR-SCA-001:** The system shall be capable of scaling to support additional university departments, laboratories, and user accounts without requiring structural changes to the core database architecture.

4.6 Maintainability

- **NFR-MNT-001:** The system shall employ a decoupled architecture, separating the React frontend from the Node.js/Express backend, so that individual components can be updated without breaking unrelated features.

4.7 Compatibility

- **NFR-CMP-001:** The system shall be a web-based platform that functions uniformly across all modern, standard web browsers (e.g., Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge).

5. User Requirements

5.1 Student Requirements

- Students require the ability to browse laboratories and available equipment.
- Students require access to view their own reservation history, borrowing history, and fines.

5.2 Technical Officer Requirements

- Technical Officers require the ability to review, approve, or reject reservations specifically for their assigned laboratory.
- Technical Officers require the ability to issue equipment, receive returns, and log maintenance tickets.

5.3 Administrator Requirements

- Administrators require the ability to manage the global hierarchy (Departments and Laboratories).
- Administrators require system-wide monitoring capabilities for equipment, reservations, and fine payments.

6. Data Requirements

6.1 Data Entities

The system shall manage the following logical data entities to support its operations:

- **User Profiles:** Storing account details, credentials, and role assignments (Student, Technical Officer, Admin).
- **Organizational Units:** Defining the university hierarchy, including specific departments and their associated laboratories.
- **Inventory Records:** Tracking equipment categories, individual equipment items, and total versus available stock quantities.
- **Transaction Records:** Storing data for student reservations, active borrowing sessions, and return logs.
- **Administrative Records:** Tracking maintenance tickets for faulty items, overdue fine calculations, and payment receipts.

6.2 Data Validation Rules

- **Role Constraints:** A user account must strictly hold one of three distinct roles, and authorization must be validated against this role for all restricted actions.

- **Reservation Logic:** The end time of an equipment reservation must always be strictly after the start time.
- **Quantity Validation:** The system shall reject any reservation request where the requested item quantity exceeds the currently available stock for that specific time.
- **State Transitions:** Reservations must follow a strict lifecycle (e.g., a reservation cannot be marked as 'Completed' unless it was first 'Approved').

6.3 Data Integrity Requirements

- **Referential Integrity:** The system must maintain strict relational links between entities. If a parent record (such as a laboratory or an equipment category) is removed, the system must safely handle dependent records (such as reservations or inventory) to prevent orphaned data.
- **Stock Accuracy:** The system shall dynamically calculate available equipment quantities to guarantee that stock levels never reflect a negative value, accounting for items that are reserved, issued, or under maintenance.

7. External Interface Requirements

7.1 User Interface

- The system shall provide a responsive, web-based graphical user interface accessible via standard web browsers.
- The interface shall adapt its navigation and dashboard features dynamically based on the authenticated user's role (Student, Technical Officer, or Admin).
- Forms for reservations, inventory updates, and ticket creation shall provide immediate, clear visual feedback and error validation to the user.

7.2 Software Interfaces

- **Payment Gateway API:** The system shall interface with the Stripe Sandbox API to facilitate the simulated processing of student fines and capture transaction reference IDs upon success.
- **Database Interface:** The backend system shall interface securely with a relational database management system to ensure persistent, structured data storage.

7.3 Communication Interfaces

- Client-server communication shall occur over standard HTTP/HTTPS protocols.
- The frontend and backend shall communicate via RESTful API endpoints using JSON for data exchange.

8. Business Rules

- **BR-001 (Equipment Issuance):** Physical equipment shall only be issued to a student, and a borrowing record created, if the corresponding reservation has been explicitly reviewed and marked as approved by a Technical Officer.
- **BR-002 (Fine Calculation):** A late penalty shall solely be applied for each full 24-hour period an item is kept past its due time; overdue durations of less than 24 hours shall not generate a fine.
- **BR-003 (Maintenance Deductions):** Any equipment unit associated with an open maintenance ticket shall be automatically and immediately deducted from the pool of available stock until the ticket is officially resolved.
- **BR-004 (Jurisdictional Boundaries):** Technical Officers shall be strictly restricted to managing reservations, inventory, and maintenance tasks only for the specific laboratory to which an Administrator has assigned them.